torch.sort#

https://pytorch.org/docs/stable/generated/torch.sort.html

Description:

Sorts the elements of the input tensor along a given dimension in ascending order by value. If dim is not given, the last dimension of the input is chosen. If descending is True then the elements are sorted in descending order by value. If stable is True then the sorting routine becomes stable, preserving the order of equivalent elements. A namedtuple of (values, indices) is returned, where the values are the sorted values and indices are the indices of the elements in the original input tensor.

Function Signature

```
torch.sort(input, dim=-1, descending=False, *, stable=False,
out=None)#

Parameters

Keyword Arguments
```

Parameters

Keyword

stable (bool, optional) – makes the sorting routine stable, which guarantees that the order of equivalent elements is preserved. out (tuple, optional) – the output tuple of (Tensor, LongTensor) that can be optionally given to be used as output buffers

Code Examples

```
>>> x = torch.randn(3, 4)
>>> sorted, indices = torch.sort(x)
>>> sorted
tensor([[-0.2162,  0.0608,  0.6719,  2.3332],
        [-0.5793,  0.0061,  0.6058,  0.9497],
        [-0.5071,  0.3343,  0.9553,  1.0960]])
>>> indices
tensor([[ 1,  0,  2,  3],
        [ 3,  1,  0,  2],
        [ 0,  3,  1,  2]])

>>> sorted, indices = torch.sort(x, 0)
>>> sorted
tensor([[-0.5071, -0.2162,  0.6719, -0.5793],
        [ 0.0608,  0.0061,  0.9497,  0.3343],
        [ 0.6058,  0.9553,  1.0960,  2.3332]])
>>> indices
tensor([[ 2,  0,  0,  1],
        [ 0,  1,  1,  2],
        [ 1,  2,  2,  0]])
>>> x = torch.tensor([0, 1] * 9)
>>> x.sort()
torch.return_types.sort(
    values=tensor([0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1,
1, 1, 1]),
    indices=tensor([ 2, 16,  4,  6, 14,  8,  0, 10, 12,  9, 17,
15, 13, 11,  7,  5,  3,  1]))
>>> x.sort(stable=True)
torch.return_types.sort(
    values=tensor([0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1,
1, 1, 1]),
    indices=tensor([ 0,  2,  4,  6,  8, 10, 12, 14, 16,  1,  3,
5,  7,  9, 11, 13, 15, 17]))
```

Detailed Documentation

Documentation

Sorts the elements of the input tensor along a given dimension in ascending order by value.

If dim is not given, the last dimension of the input is chosen.

If descending is True then the elements are sorted in descending order by value.

If stable is True then the sorting routine becomes stable, preserving the order of equivalent elements.

A namedtuple of (values, indices) is returned, where the values are the sorted values and indices are the indices of the elements in the original input tensor.

input (Tensor) – the input tensor.

dim (int, optional) – the dimension to sort along

descending (bool, optional) – controls the sorting order (ascending or descending)

stable (bool, optional) – makes the sorting routine stable, which guarantees that the order of equivalent elements is preserved.

out (tuple, optional) – the output tuple of (Tensor, LongTensor) that can be optionally given to be used as output buffers